



AI/LLM Application Builder From Zero to Mastery

A practice-first industrial training in modern AI application engineering — one real system, built session by session, deployed live.

 8 Weeks

 16 Sessions

 32 Contact Hours

 Friday & Saturday · 7:30–9:30 PM

 Live · Instructor-led

 Prepared for the **Department of Computer & Communication Engineering** — as part of the students' industrial training program.

Program overview

This training bridges the gap between academic coursework and industry AI engineering practice. Instead of surveying disconnected topics, participants spend all sixteen sessions building **one production-style AI system** — the way a real engineering team would: incrementally, with version control, testing, observability, and a final live deployment. Every session ends with the project visibly more capable than it began, and every version is committed so students can trace exactly what they built and why. The program uses the current industry stack — Python, LangChain & LangGraph, retrieval (RAG), FastAPI, React, and AI-assisted development.

The training project — “ShopWise”, an AI service desk for an online store

A typical e-commerce shop drowns in support messages: “Where is my order?” · “Can I return this?” · “I was charged twice!” Students build the complete AI system that runs this shop's service operation:

WHO IT SERVES



Customers

A 24/7 support chat that checks real orders & delivery status, explains return and warranty policy from the company's actual documents, and hands over to a human when it should.



Support manager

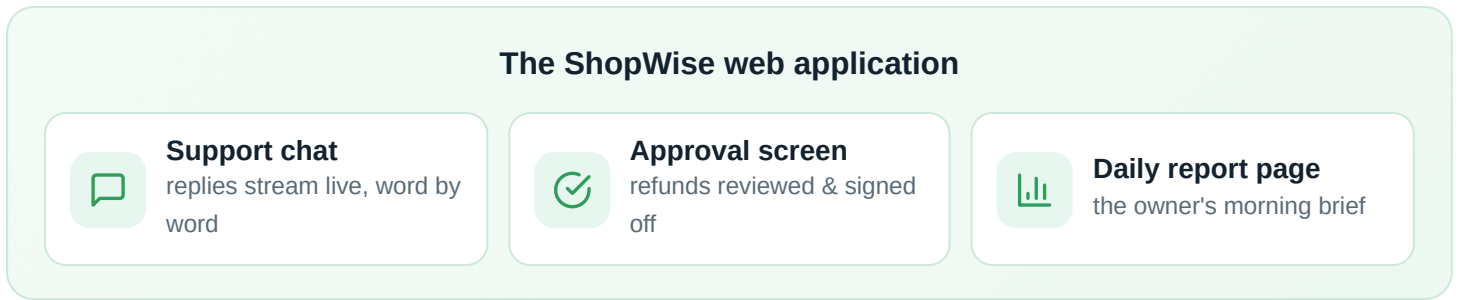
Nothing risky happens alone: every refund waits for the manager's approve / adjust / reject — from any device, with a full audit trail.



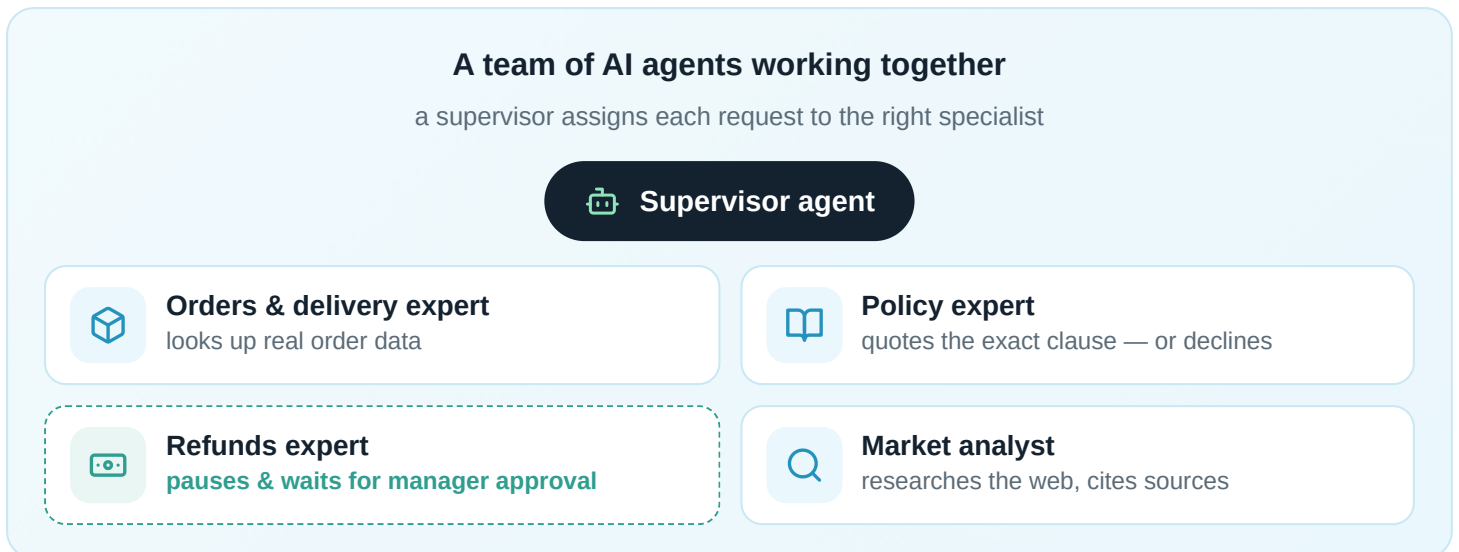
Shop owner

A one-page business brief every morning — sales, refunds, complaints — plus live market and competitor price research.

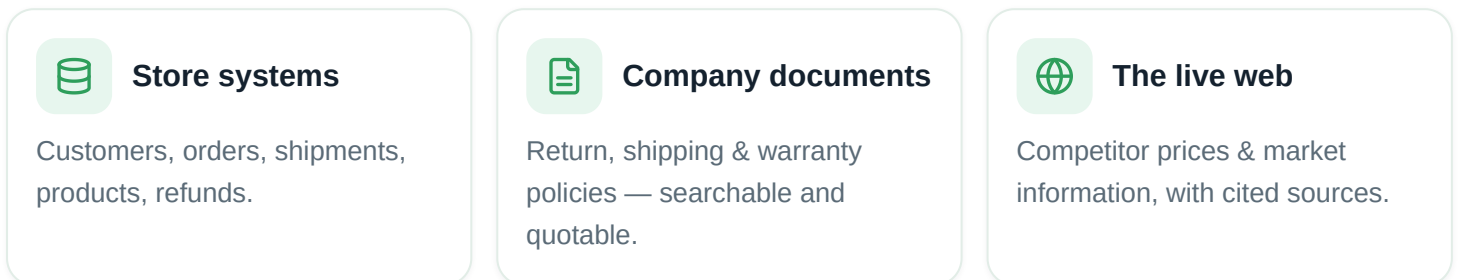
THE APPLICATION STUDENTS BUILD



THE INTELLIGENCE BEHIND IT



WHAT IT DRAWS ON



Under the hood — the industry engineering practices students implement to make this real:

- ✓ Tool-using AI agents grounded in real data (never inventing facts)
- ✓ Document search & retrieval (RAG) with citations and honest refusals
- ✓ Human-in-the-loop approval workflows that pause and resume
- ✓ Conversation memory per ticket + durable customer profiles
- ✓ Safety engineering: prompt-injection filters, PII redaction, cost limits
- ✓ Multi-agent orchestration with explicit, auditable workflows
- ✓ Full observability — every AI decision traceable and debuggable
- ✓ Web API + modern web interface + cloud deployment & scheduling

Session-by-session plan

Wk	S#	Session focus	The project after this session
1	S1	How AI language models actually work — tokens, context, why they hallucinate; environment setup	Foundations laid; tools installed and verified
	S2	Talking to LLMs — messages, roles, and system prompts that control behavior	First automated support reply, in the company's voice
2	S3	Outputs software can trust — structured data, error handling, cost tracking	Tickets triaged into category, urgency & sentiment
	S4	LangChain: one interface for every AI provider + LangSmith observability	Vendor-independent pipeline, every call traceable
3	S5	Tools & agents — giving the AI “hands” to look things up	Answers from real order, shipping & product data
	S6	Conversation memory & live streaming replies	Multi-turn support chat that remembers the ticket
4	S7	Safety rails — guardrails, PII protection, human approval on risky actions	Refunds now require a manager's sign-off
	S8	Teaching it the company's documents — indexing & semantic search (RAG part 1)	Policy documents searchable by meaning
5	S9	Grounded answers — citations and honest refusals (RAG part 2)	Policy answers quote the exact clause — or decline
	S10	LangGraph — turning the AI into an explicit, auditable workflow	Enforced process: triage → route → resolve
6	S11	Pause for people; memory that lasts — approvals & customer profiles	Approvals resumable days later; customers remembered across tickets
	S12	A team of AI specialists — multi-agent architecture	Supervisor delegating to order / policy / refund experts
7	S13	FastAPI backend — serving the assistant to real users	Web API with streaming chat & approval endpoints
	S14	A modern web interface — built with AI pair-programming (Claude Code)	Full web app: chat, approval screen, report page

Wk	S#	Session focus	The project after this session
8	S15	Cloud deployment & automation — going live, scheduled daily reports	Running on the internet with durable memory
	S16	Final presentations — students demo their own extensions; evaluation	Sixteen personalized capstone versions, defended live

Training outcomes

On completion, participants are able to:

- ✓ Design and implement AI/LLM applications end-to-end — agents, retrieval, memory, safety, and human oversight
- ✓ Serve and deploy an AI system as a real product (API, web UI, cloud)
- ✓ Debug AI behavior systematically from execution traces
- ✓ Apply industry engineering discipline: versioning, testing, cost control, observability
- ✓ Present and defend a working system — each student demos a personalized capstone

Assessment & materials

- ✓ Graded exercises with solutions in every session
- ✓ Continuous project milestones (16 committed versions)
- ✓ Rubric-based capstone evaluation with a final demo day
- ✓ All materials provided: runnable notebooks, project repository, an advanced self-study track, and an interview-preparation question bank
- ✓ Certificate of completion issued by Grow with Data

Prerequisites

- ▶ Programming fundamentals & basic Python
- ▶ A beginner's familiarity with AI/LLMs (having used ChatGPT is sufficient)
- ▶ Own laptop — all tools and AI models used are free-tier; no cost to students

Conducted by Grow with Data

Grow with Data delivers industry-grade, hands-on AI engineering training, designed and taught by practitioners who ship AI products in production for the US market.

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